Hyperspectral Reconstruction From RGB Images via Physically Guided Graph Deep Prior Learning

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***Abstract*— Recovering the latent hyperspectral image (HSI) from RGB or multispectral image (MSI), which is dubbed spectral super-resolution (SSR), has demonstrated outstanding performance owing to the advancements in convolutional neural networks (CNNs). However, most of the current algorithms concentrate on the pursuit of networks with more expensive or complex structures, while ignoring the significant role of physical degradation models in SSR. In addition, the inherent defects of CNN make these networks focus more on the local correlation, while their ability to model the long-range correlations in the spectral and spatial domains still has room to improve. To overcome this shortcoming, we propose a physical degradation-guided deep prior learning network (PGDL-Net) for SSR via unfolding the optimization process of the blind SSR model, in which the priors of unknown spectral response function (SRF) and latent HSI are learned explicitly and represented by proximal operators. To jointly extract the local and non-local information, we design a hybrid graph Transformer as the proximal operator to solve the latent HSI. Furthermore, to ensure efficient learning of SRF and HSI, we also propose a novel loss function constraining the reconstruction error, degradation consistency, and observation fidelity for the learned SRF and HSI. Experimental results on multiple datasets illustrate the improved performance and stability of our method in SSR.**

***Index Terms*— Deep prior, degradation, proximal operator, spectral super-resolution (SSR).**

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I. Introduction

HYPERSPECTRAL imaging records hundreds of detailed, continuous, and narrow-band spectra of a scene along the spectral axis. We are witnessing increasing demands and applications of hyperspectral image (HSI) in numerous domains, such as visual tracking [1], [2], [3], remote sensing [4], [5], [6], and classification [7], [8], [9]. In comparison to RGB images, HSIs possess a higher quantity of spectral bands, allowing for the preservation of more abundant information and the depiction of finer details of the captured scenes. Nevertheless, the practical use of HSIs is often restricted by hardware limitations, including the ineluctable compromise between spectral and spatial resolutions, which means that the spatial resolution has to be reduced for a higher spectral resolution [10]. These limitations set an insurmountable barrier for high-resolution HSIs and hinder their potential applications. Some scholars blend on acquiring HSI solely from corresponding RGB images, which is referred to as spectral super-resolution (SSR) [11], [12], [13], [14], [15], [16], [17], [18].

Conventional SSR methods mainly concentrate on sparse coding [19]. The objective of SSR using sparse representation is to extract the over-complete hyperspectral dictionary that covers all hyperspectral pixels [20]. However, the model-based methods [21], [22], [23] are usually shallow models and suffer from poor generalization and representation capabilities. Attributed to recent developments in deep learning, SSR has made significant advancements. The mapping function between RGB image and HSI, in particular, can be learned with the help of deep convolutional neural networks (CNNs) [14], [24], [25], [26].

Despite achieving impressive results, most of the current methods ignore the physical degradation process of HSIs. The spectral response function (SRF) records the sensor's response to different wavelengths [27]. It plays an important role in hyperspectral imaging, while there has been few research explicitly exploiting the SRF in SSR. Although some other deep unfolding SSR networks, such as [28], integrate the spectral degradation by representing the SRF with a 1×1 learnable *conv.* layer, the learned SRF is actually an "average" distribution of all training samples. The generative ability to different SRFs, particularly unseen SRFs, is limited. On the

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other hand, the long-range correlations between spectra are important characteristics of HSI. Transformers with multihead self-attention (MSA) [29], [30] perform well in modeling the long-range correlations. But how to better exploit the local and non-local dependencies in HSI still remains an open issue and has room to improve.

To address the aforementioned limitations, by explicitly learning the priors of SRF and latent HSI, and fully exploiting the local and non-local correlations in spectral–spatial dimensions, we propose a physical degradation-guided deep prior learning network (PGDL-Net) for SSR via unfolding the optimization process of blind SSR model, in which both the priors of unknown SRF and latent HSI are jointly learned. To fully extract the local and non-local information simultaneously, we propose a hybrid graph Transformer as the deep proximal operator of HSI. Furthermore, a novel loss function is designed for constraining the reconstruction error, degradation consistency, and observation fidelity for the learning of SRF and HSI.

The main contributions can be summarized as follows.

1) We propose a blind SSR model by fully exploiting the physical degradation model. The unknown SRF and latent HSI are jointly optimized and iteratively solved in our joint SSR model. By unfolding the optimization process, the HSI and SRF are jointly learned by two types of deep proximal operators, thus both HSI and SRF can be explicitly inferred, which differs from the other deep unfolding networks.
2) We propose a hybrid graph Transformer as a proximal operator in the SSR network to represent the deep prior of HSI. The proposed hybrid graph Transformer could not only explore the long-range dependence as other Transformers but also could focus more on the local contextual correlation via the neighboring aggregation of graph convolutional network (GCN).
3) To enhance the learning of SRF, we design a novel loss function, which is composed of three parts, including HSI reconstruction loss, degradation consistency loss ($\mathcal{L}_{\text{dc}}$), and observation fidelity loss ($\mathcal{L}_{\text{of}}$). The experiments demonstrate that the proposed loss benefits the SSR performance.

II. Related Works

Traditional SSR methods mainly work on exploiting the hand-crafted priors. For instance, Parmar et al. [20] proposed a sparsity prior-based method to reconstruct HSI. Instead of reconstructing each spectrum independently, it jointly reconstructed the spatial–spectral data cube by exploiting the sparsity prior. Arad and Ben-Shahar [19] suggested learning a mapping between HSI and RGB via sparse coding for SSR, which utilizes the sparsity prior to generating a sparse dictionary between HSI and the corresponding RGB image. Akhtar et al. [22] proposed a non-parametric Bayesian sparse representation, which integrated high-resolution images with low-resolution HSIs, acquiring HSIs free from physical device restriction. However, these model-based methods are always subjected to poor generalization and representation capabilities, making them inappropriate for more complicated scenarios.

Recently, owing to the outstanding performance of deep learning in computer vision [31], [32], [33], many scholars have attempted to leverage CNN in SSR. For example, Zhao et al. [12] proposed a four-level regression network using a pixel shuffle to connect layers of different levels, which considered context information explicitly and ensured the information fidelity. Shi et al. [13] proposed hyperspectral recovery CNN (HSCNN), a deep network built on residual blocks which has two types of framework, HSCNN-densenet (HSCNN-D) and HSCNN-resnet (HSCNN-R), which substantially deepened the network for an accurate solution. Li et al. [14] presented a hybrid attention CNN using a structural consistency mechanism, which was stacked with several residual channels and bands attention modules, enhancing the channel extraction and fusion performance.

Attribute to the advantage of obtaining long-range correlations in the spatial and spectral domains, the Transformer model has been employed in low-level vision [34], [35], [36], [37] and has also been employed in HSI reconstruction and acquired a lot of interest [38], [39], [40]. For instance, Cai et al. [40] proposed a Transformer-stacked end-to-end framework, dubbed multi-stage spectral-wise Transformer (MST++), for SSR, which used the MSA in the spectral domain, showing superiority in obtaining the long-range correlations between spectra. Wang et al. [41] proposed a novel hybrid spectral-wise attention Transformer, which can simultaneously model the spectral-wise similarity and particularity.

Model-guided deep unfolding network for HSI reconstruction has been widely explored [42], [43], [44], [45], [46]. For example, Dian et al. [43] presented SSRNet, a cross-fusion-based model-guided network, which can learn the spatial and spectral priors of HSI. Li et al. [44] proposed a pixel adaptive deep unfolding Transformer and introduced a novel frequency perspective using amplitude and phase representations to interact between cross-stage features. Some deep unfolding networks employ Transformer as proximal operators. Cai et al. [45] unfolded the reconstruction model of HSI into the deep network and proposed a half-shuffle Transformer as the deep proximal operator for local contexture and non-local dependency. Ma et al. [46] proposed a novel super-resolution method by leveraging the Transformer to learn the prior of HSIs and adding a 3D-CNN after the Transformer to capture the spatial–spectral correlation of HSI. Generally, the efficiency of the Transformer in capturing local and non-local information in the spatial domain can still be improved.

These deep unfolding networks implicitly represent the degradation using learnable network parameters, but they could only estimate a single SRF, which is actually the average of the training samples. Comparatively, the proposed method represents the unknown SRF through deep proximal learning, which endorses it the ability to infer different SRFs, particularly unseen SRFs, during the inference.

III. Proposed PGDL-Net for SSR

A. Model Formulation

In blind HSI SSR, both the latent HSI and SRF are unknown variables and need to be solved. In this study, on the basis of the spectral relationship between HSI and RGB images,

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we propose a joint SSR model, in which both latent HSI and SRF can be estimated explicitly.

Specifically, let the latent HSI represented by $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, where C is the number of spectral bands, and H and W are height and width, respectively. $\mathbf{Y} \in \mathbb{R}^{c\times H\times W}$ denotes the input RGB image that only has c bands but is consistent with $\mathbf{X}$ in spatial resolution. $\mathbf{R} \in \mathbb{R}^{c\times C}$ is the SRF for the spectral degradation process between HSI and RGB and $\mathbf{N}$ denotes the additive noise. The observation model is

$$\mathbf{Y} = \mathbf{R}\mathbf{X} + \mathbf{N}. \tag{1}$$

To solve this ill-posed problem, we formulate it as an optimization problem of minimizing an energy function with prior knowledge imposed on both unknown SRF and HSI, the energy function can be written as

$$\hat{X}, \hat{R} = \underset{\mathbf{X},\mathbf{R}}{\operatorname{argmin}} \frac{1}{2}\left\|\mathbf{Y} - \mathbf{R}\mathbf{X}\right\|_F^2 + \lambda_1 J_1(\mathbf{X}) + \lambda_2 J_2(\mathbf{R}) \tag{2}$$

where $J_1(\cdot)$ and $J_2(\cdot)$ represent the regularization terms that impose the prior knowledge. In this study, we propose to use deep prior J_{deep} to regularize SRF and HSI, due to the fact that the deep prior can be learned from data flexibly. $\|\cdot\|_F^2$ is the Frobenius norm of the data-fidelity term. λ_1 and λ_2 are trade-off parameters among data-fidelity and regularization terms. This energy function can guarantee the consistency between the observed RGB and the reconstructed HSI. To solve this optimization problem, by introducing the auxiliary variables $\mathbf{V}$, $\mathbf{K}$, we separate the data-fidelity term from the regularization term, making it easier to solve. The optimization problem can be rewritten as

$$\begin{aligned} \hat{X}, \hat{R} = & \underset{\mathbf{X},\mathbf{R}}{\operatorname{argmin}} \frac{1}{2}\left\|\mathbf{Y} - \mathbf{K}\mathbf{V}\right\|_F^2 + \lambda_1 J_1(\mathbf{X}) + \lambda_2 J_2(\mathbf{R}) \\ & \text{s.t.} \quad \mathbf{V} = \mathbf{X}, \mathbf{K} = \mathbf{R}. \end{aligned} \tag{3}$$

By optimizing the energy function above, we can get solutions for the SRF and HSI. It is challenging to handle this issue directly. To solve (3), we split it into the $\mathbf{X}$- and $\mathbf{R}$-subproblems

$$\hat{X} = \underset{\mathbf{X}}{\operatorname{argmin}} \frac{1}{2}\left\|\mathbf{Y} - \mathbf{K}\mathbf{V}\right\|_F^2 + \mu_1\left\|\mathbf{X} - \mathbf{V}\right\|_F^2 + \lambda_1 J_{\text{deep}}(\mathbf{X}) \tag{4}$$

$$\hat{R} = \underset{\mathbf{R}}{\operatorname{argmin}} \frac{1}{2}\left\|\mathbf{Y} - \mathbf{K}\mathbf{V}\right\|_F^2 + \mu_2\left\|\mathbf{R} - \mathbf{K}\right\|_F^2 + \lambda_2 J_{\text{deep}}(\mathbf{R}) \tag{5}$$

where μ_1 and μ_2 are penalty parameters. By using the half-quadratic splitting method, four subproblems that split from the optimization problems could be solved more quickly and effectively

$$\hat{V} = \underset{\mathbf{V}}{\operatorname{argmin}} \frac{1}{2}\left\|\mathbf{Y} - \mathbf{K}\mathbf{V}\right\|_F^2 + \mu_1\left\|\mathbf{X} - \mathbf{V}\right\|_F^2 \tag{6}$$

$$\hat{X} = \underset{\mathbf{X}}{\operatorname{argmin}} \frac{1}{2}\left\|\mathbf{V} - \mathbf{X}\right\|_F^2 + \frac{\lambda_1}{\mu_1} J_{\text{deep}}(\mathbf{X}) \tag{7}$$

$$\hat{K} = \underset{\mathbf{K}}{\operatorname{argmin}} \frac{1}{2}\left\|\mathbf{Y} - \mathbf{K}\mathbf{V}\right\|_F^2 + \mu_2\left\|\mathbf{R} - \mathbf{K}\right\|_F^2 \tag{8}$$

Algorithm 1 Unfolding Process of PGDL-Net

Input: RGB $\mathbf{Y}$, initial SRF $\mathbf{R}$
Output: HSI $\mathbf{X}$, optimized SRF $\mathbf{R}'$
Initialization : $\mathbf{X}_0$, $\mathbf{R}_0$, $\mathbf{Y}_0$, $\mathbf{V}_0$, $\mathbf{K}_0$
while $s < S$ **do**
  Solving $\mathbf{K}_s$ via Eq. (11);
  Solving $\mathbf{V}_s$ via Eq. (10);
  $\mathbf{Y}_s = conv_{3\times3}(\mathbf{Y}_{s-1})$;
  Solving $\mathbf{R}_s$ via Eq. (13);
  Solving $\mathbf{X}_s$ via Eq. (12);
end
$\mathbf{X} \leftarrow \mathbf{X}_S$, $\mathbf{R} \leftarrow \mathbf{R}_S$

$$\hat{R} = \underset{\mathbf{R}}{\operatorname{argmin}} \frac{1}{2}\left\|\mathbf{K} - \mathbf{R}\right\|_F^2 + \frac{\lambda_2}{\mu_2} J_{\text{deep}}(\mathbf{R}). \tag{9}$$

The $\mathbf{V}$- and $\mathbf{K}$-subproblems can be optimized by the gradient descent algorithm. The solutions at the sth iterative stage are

$$\begin{aligned} \mathbf{V}_s = & \ \mathbf{V}_{s-1} - \epsilon_1(\mathbf{K}_{s-1}^T\mathbf{K}_{s-1}\mathbf{V}_{s-1} - \mathbf{K}_{s-1}^T\mathbf{Y}_{s-1} \\ & + \mu_1\mathbf{V}_{s-1} - \mu_1\mathbf{X}_{s-1}) \\ = & \ (1-\epsilon_1\mu_1)\mathbf{V}_{s-1} + \epsilon_1\mu_1\mathbf{X}_{s-1} + \epsilon_1\mathbf{K}_{s-1}^T\mathbf{Y}_{s-1} \\ & - \epsilon_1\mathbf{K}_{s-1}^T\mathbf{K}_{s-1}\mathbf{V}_{s-1} \end{aligned} \tag{10}$$

$$\begin{aligned} \mathbf{K}_s = & \ \mathbf{K}_{s-1} - \epsilon_2(\mathbf{K}_{s-1}\mathbf{V}_{s-1}\mathbf{V}_{s-1}^T - \mathbf{V}_{s-1}\mathbf{Y}_{s-1}^T \\ & + \mu_2\mathbf{R}_{s-1} - \mu_2\mathbf{K}_{s-1}) \\ = & \ (1-\epsilon_2\mu_2)\mathbf{K}_{s-1} + \epsilon_2\mu_2\mathbf{R}_{s-1} + \epsilon_2\mathbf{R}_{s-1}\mathbf{V}_{s-1}^T \\ & - \epsilon_2\mathbf{K}_{s-1}\mathbf{V}_{s-1}\mathbf{V}_{s-1}^T \end{aligned} \tag{11}$$

where ϵ_1 and ϵ_2 are the step sizes. For simplicity, there is only one step in (10) and (11). The $\mathbf{X}$- and $\mathbf{R}$-subproblems are solved by the proximal operators representing the deep prior learning

$$\mathbf{X}_s = \text{Prox}_{\lambda_1/\mu_1}(\mathbf{V}_s) \tag{12}$$

$$\mathbf{R}_s = \text{Prox}_{\lambda_2/\mu_2}(\mathbf{K}_s). \tag{13}$$

The aforementioned four subproblems can be optimized alternately to solve the optimization problem. The overall optimization of our joint SSR model is shown in Algorithm 1.

B. Network Architecture

As mentioned in Section III-A, the SSR problem can be solved by optimizing (6)–(9). However, updating the four equations alternately needs manual intervention, which impacts the efficiency and quality of SSR.

In this study, we unfold the optimization algorithm into a learnable end-to-end network. By this means, the latent HSI and unknown SRF will be learned with the elaborately designed HSI proximal operator and the SRF proximal operator. The overall network structure of the proposed algorithm is depicted in Fig. 1.

Given $\mathbf{Y}$, $\mathbf{R}$, the shallow features $\mathbf{Y}_0$ and $\mathbf{R}_0$ are first generated by the initialization block, which consists of convolutional layers with an activation function

$$\mathbf{Y}_0 = \text{ReLU}\left(\mathbf{W}_0^Y * \mathbf{Y} + b_0^Y\right) \tag{14}$$

$$\mathbf{R}_0 = \text{ReLU}\left(\mathbf{W}_0^R * \mathbf{R} + b_0^R\right) \tag{15}$$

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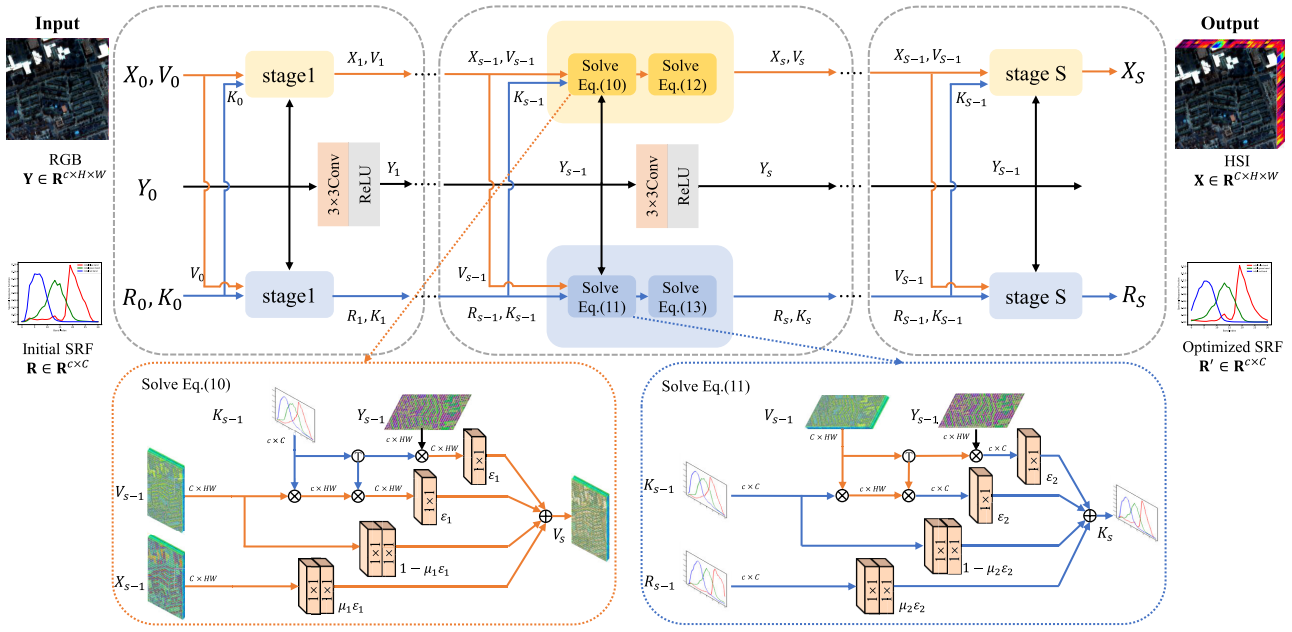


Fig. 1. General architecture of the proposed PGDL-Net for SSR. It takes an RGB image and an initial SRF as input, then estimates the latent HSI and SRF. There are several iterative stages, the unfolding processes of solving (10) and (11) are given as well, and the deep proximal operators of (12) and (13) are presented in Fig. 2.

where $\mathbf{W}_0^Y$ and b_0^Y are trainable kernel weights and biases for the convolutional layer of RGB image in initialization, and $\mathbf{W}_0^R$ and b_0^R are that of SRF. Then, to generate the initialization of HSI, a convolutional layer is employed

$$\mathbf{X}_0 = \mathrm{ReLU}\left(\mathbf{W}_0^X * \mathbf{Y}_0 \ + \ b_0^X\right). \tag{16}$$

Let $\mathbf{V}_0 = \mathbf{X}_0$ and $\mathbf{K}_0 = \mathbf{R}_0$, the initialized features $\mathbf{X}_0$, $\mathbf{R}_0$, $\mathbf{V}_0$, $\mathbf{K}_0$, and $\mathbf{Y}_0$ will be fed into the unfolding stages to acquire the ultimate optimized outcomes stage by stage. In the sth stage, these variables are solved using the learnable network parameters, each of them corresponds to (10)–(13), as shown in Fig. 1. As for $\mathbf{K}$- and $\mathbf{V}$-subproblems, we fuse the feature maps of the previous stage according to (10) and (11). To learn more deep features, a convolutional layer is employed to obtain deeper features of $\mathbf{Y}_s$. It is not mathematically necessary to update $\mathbf{Y}_s$ in the iterative process. But in the deep unfolding network, the variables in each iterative solution would be the learned deep features, and we replace the fixed $\mathbf{Y}$ with an iterative updated $\mathbf{Y}_s$ through a *conv.* layer. Such design benefits learning deeper features from $\mathbf{Y}_s$ and making the network more flexible. We set the penalty parameters and step size as learnable parameters and use 1×1 *conv.* layers to learning them.

The HSI prior $J_{\text{deep}}(\mathbf{X})$ and the SRF prior $J_{\text{deep}}(\mathbf{R})$ can be learned by the proximal operators implicitly, thereby providing a practical and efficient approach to iteratively enhance and optimize the spectral recovery process. The detailed structure of proximal operators is presented in Section III-C.

C. Proximal Operators

1) Operator for $\mathbf{R}$*-Subproblem:* Similar to [42], we employ the fully connected (FC) layers to design the proximal operator for the $\mathbf{R}$-subproblem, as shown in Fig. 2(d), since the spectral degradation $\mathbf{R}_s$ can be thought of as a vector with length cC. With the help of the deep proximal operator, the SRF $\mathbf{R}_s$ can be reconstructed from the auxiliary variable $\mathbf{K}_s$. We first vectorize the input matrix $\mathbf{K}_s$ and then feed it into the FC layers. Each of the two hidden layers has 64 neurons, and there are cC neurons in both the input and output layers.

2) Operator for $\mathbf{X}$*-Subproblem:* The learning-based algorithms can explicitly represent the prior, which is hard to be completely captured by the traditional hand-crafted priors. Previous deep unfolding networks usually employ the CNN as the proximal operator. However, CNN methods focus on extracting the local feature representations through convolutional layers, losing sight of non-local spatial information and long-range correlations along the spectral domain. Transformer can extract context-sensitive feature representations through an MSA mechanism and model the long-range correlations, and GCN could effectively extract the non-local features in the spatial domain [47], [48], [49], [50], [51], [52]. To solve the $\mathbf{X}$-subproblem in (12), the proposed method integrates CNN, GCN, and Transformer to constitute a hybrid graph Transformer as the deep proximal operator of HSI for modeling the local correlation and long-range dependence in the spectral and spatial domains. As shown in Fig. 2(a), we design a subnetwork with a two-branch structure combining the superiorities of CNN, Transformer, and GCN that extracts the local and non-local information simultaneously. The subnetwork is made up of a spatial information sub-network (SISn) for modeling the local spatial information, a global transformer sub-network (GTSn) for modeling the long-range correlations between spectra, and a pixel-level graph sub-network (PGSn) for the non-local spatial information.

a) Spatial information sub-network: To exploit the local correlation, we employ two convolutional layers with the nonlinear activation unit rectified linear unit (ReLU) in our

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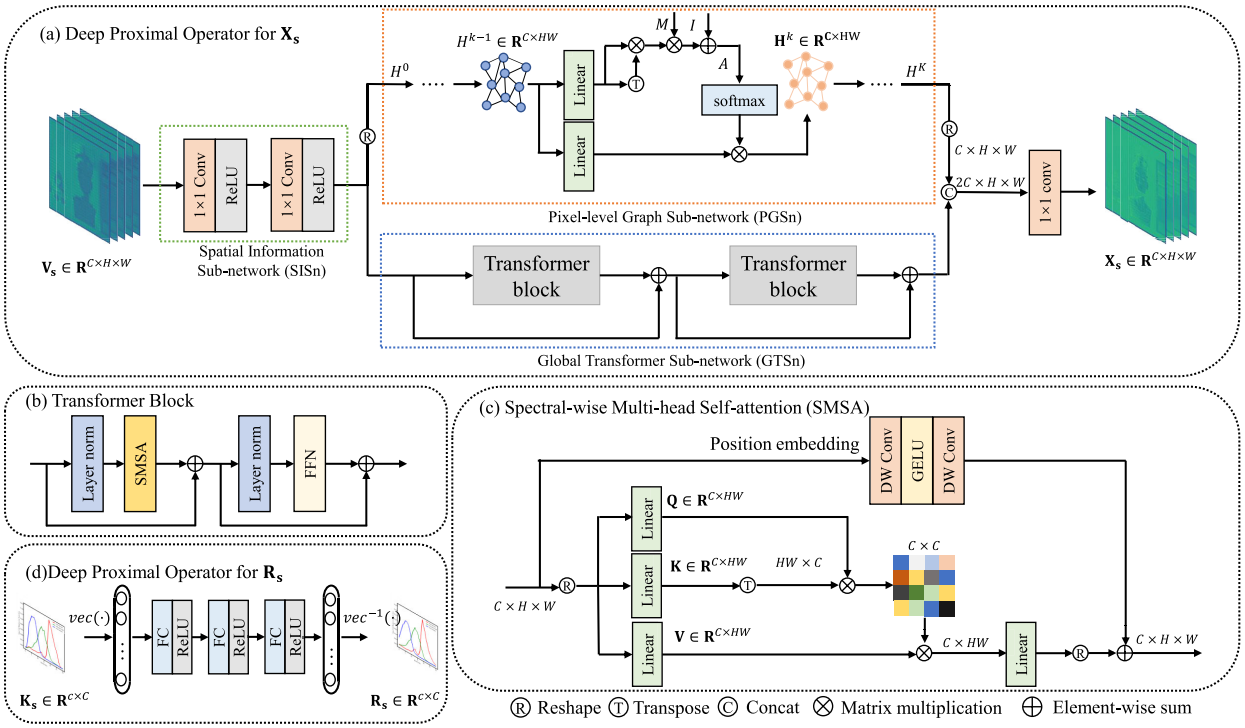


Fig. 2. Deep proximal operators for HSI and SRF. (a) Proposed hybrid graph transformer as the deep proximal operator of HSI. (b) Structure of the transformer block. (c) Spectral-wise MSA. (d) Network of the deep proximal operator of SRF.

SISn. The feature maps $\mathbf{V}_s$ will be fed into this sub-network to model the local spatial and spectral information.

b) Global transformer sub-network: The GTSn is composed of two spectral-wise attention blocks and skip connections. As shown in Fig. 2(b), each block is made up of a spectral-wise MSA, two normalization layers, and a feed-forward network. Let $\mathbf{X}_{\text{IN}} \in \mathbb{R}^{C\times H\times W}$ represent the input of MSA, which is then transformed into tokens $\mathbf{X} \in \mathbb{R}^{C\times HW}$. Subsequently, $\mathbf{X}$ will be mapped to query $\mathbf{Q} \in \mathbb{R}^{C\times HW}$, key $\mathbf{K} \in \mathbb{R}^{C\times HW}$, and value $\mathbf{V} \in \mathbb{R}^{C\times HW}$

$$\mathbf{Q} = \mathbf{W}^Q\mathbf{X}, \mathbf{K} = \mathbf{W}^K\mathbf{X}, \mathbf{V} = \mathbf{W}^V\mathbf{X} \tag{17}$$

where $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V \in \mathbb{R}^{C\times C}$ are weight matrices, and bias is omitted for simplification. Afterward, let $d_h = C/N$ be the size of each head, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are then divided into multiple heads along the channel axis: $\mathbf{Q} = [\mathbf{Q}^1, \mathbf{Q}^2, \ldots, \mathbf{Q}^N]$, $\mathbf{K} = [\mathbf{K}^1, \mathbf{K}^2, \ldots, \mathbf{K}^N]$, $\mathbf{V} = [\mathbf{V}^1, \mathbf{V}^2, \ldots, \mathbf{V}^N]$. For each attention head, the attention matrix $\mathbf{A}_i$ and attention value v are computed as

$$\mathbf{A}_i = \text{softmax}\left(\frac{\mathbf{Q}_i\mathbf{K}_i^T}{\sqrt{d_h}}\right), \quad v_i = \mathbf{A}_i\mathbf{V}_i \tag{18}$$

where d_h is a scaling factor for the attention scores. Due to the significant variation of the spectral curve across wavelengths, the scaling factor can re-weight the matrix, helping align the calculation result with the physical reality. Then, the context feature matrices will be concatenated along the spectral dimension

$$\text{Output} = \mathbf{W}(\text{Concat}(v_i)) + p(\mathbf{X}) \tag{19}$$

where $\mathbf{W} \in \mathbb{R}^{C\times C}$ are parameters to be learned. $p(\mathbf{X})$ generates the position information using two depth-wise convolutional layers and a Gaussian error linear unit (GELU) activation layer. The position embedding is to retain the original relative position information of pixels during the self-attention calculation. We think that the spatial position information of images is significant for SSR. After that, the final result will be reshaped into output $\mathbf{X}_{\text{OUT}} \in \mathbb{R}^{C\times H\times W}$.

c) Pixel-level graph sub-network: In the PGSn, feature maps will be first reshaped into node matrix $\mathbf{H} \in \mathbb{R}^{C\times HW}$, where each row of $\mathbf{H}$ corresponds to a specific node. Generally, it is not easy to directly merge CNN and GCN due to the mismatch in data representation structures. The GCN is designed for graphs with arbitrary structure, whereas CNN operates on data in a standard rectangular structure. Therefore, we split the whole image into small patches and then regard each individual pixel as a node within the graph to capture the global correlation in the spatial domain. Based on the input image, we initialize the adjacency correlation mask $\mathbf{M} \in \mathbb{R}^{N\times N}$ as

$$\mathbf{M}_{i,j} = \begin{cases} 1, & \text{if } p_i \text{ and } p_j \text{ are adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{20}$$

where $\mathbf{M}_{i,j}$ means whether the ith node is adjacent to the jth node, $N = HW$. We use the four-link method to calculate the mask, which means each pixel is only adjacent to pixels in four directions.

Subsequently, the nodes and edges defined on the pixel-level graph are then encoded into an adjacency weighting matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$ and node matrix $\mathbf{H} \in \mathbb{R}^{HW\times C}$. It should be noted that $\mathbf{A}_{i,j}$ refers to the correlation from the ith node to the jth node, and the ith node is defined by the ith row of $\mathbf{H}$. The initial adjacency weighting matrix $\mathbf{A}$ is numerically identical

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to $\mathbf{M}$. Shuman et al. [51] decomposed the Laplacian matrix of a graph along the spectral dimension, extending conventional convolution to the graph domain. Let $\mathbf{D}$ be the weighting degree matrix of $\mathbf{A}$, we can get the Laplacian matrix of a graph by the formulation $\mathbf{L} = \mathbf{D}^{-(1/2)}(\mathbf{D} - \mathbf{A})\mathbf{D}^{-(1/2)}$. Then, the graph convolutional layer can be defined as

$$\boldsymbol{y} = \eta\left(\mathbf{I} + \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)}\right)\boldsymbol{x} \tag{21}$$

where $\boldsymbol{y}$ and $\boldsymbol{x}$ mean the output and input vectors, $\mathbf{I}$ is the identity matrix, $\eta = 0.5$. Repeating this operation may result in unsteady estimating because the eigenvalues of $\mathbf{I} + \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)}$ are within the scope of [0, 2]. To overcome this shortcoming, Kipf and Welling [47] introduced a self-loop for each node to improve the graph convolution, which means adding an extra $\mathbf{I}$ to the matrix $\mathbf{A}$. Furthermore, Liu et al. [53] proposed to recalculate the weights of the adjacency matrix according to the output of the previous layer to enhance the adaptation of the graph convolutional layer to hyperspectral data. Let $\mathbf{H}^{k-1}$ be the input of the kth layer, then the adjacency weighting matrix of the kth layer can be rescaled as

$$\mathbf{A}^k = \text{Sigmoid}\left(\left(\mathbf{W}^k\mathbf{H}^{k-1}\right)\left(\mathbf{W}^k\mathbf{H}^{k-1}\right)^T\right) \odot \mathbf{M} + \mathbf{I} \tag{22}$$

where $\mathbf{A}^k$ is the recalculated adjacency weighting matrix in the kth layer, $\mathbf{W}^k$ is learnable parameter, $\mathbf{M}$ means the mask generated by (20), and the operator $\odot$ means the matrix multiplication. Ultimately, the graph convolutional layer is

$$\mathbf{H}^k = f\left((\mathbf{D}^k)^{-(1/2)}\mathbf{A}^k(\mathbf{D}^k)^{-(1/2)}\mathbf{W}_\varphi^k\mathbf{H}^{k-1}\right) \tag{23}$$

where $\mathbf{D}^k$ means the weighting degree matrix of $\mathbf{A}^k$ and $\mathbf{W}_\varphi^k$ represents the learnable parameters. By designing this branch, we can extract the non-local information of feature maps.

It is noted that the spatial relationship can also be exploited by applying a Transformer in the spatial domain as [38] and [39]. The proposed hybrid graph Transformer could not only explore the spatial long-range dependence as the current spatial-wise Transformers but also could focus more on the local contextual correlation via the neighboring aggregation of GCN, which benefits the HSI SSR.

D. Loss Function

Previous SSR networks usually adopt a reconstruction loss computing the disparity in pixels between the reconstructed and the ground-truth HSIs, defined as

$$\mathcal{L}_{\text{re}} = \frac{1}{N}\sum_{i=0}^{N}\left\|\hat{\mathbf{X}}^i - \mathbf{X}^i\right\|_F^2 \tag{24}$$

where $\hat{\mathbf{X}}^i \in \mathbb{R}^{C\times HW}$ represents the reconstructed HSI and $\mathbf{X}^i \in \mathbb{R}^{C\times HW}$ represents the ground truth. N means the total count of samples. However, a single reconstruction loss cannot reflect the degradation process of HSI, potentially resulting in an over-smoothed image. In addition, previous methods take SRF as known knowledge without optimization, which may lead to a performance drop in data with different SRFs. Therefore, we design two extra losses, degradation consistency loss ($\mathcal{L}_{\text{dc}}$), and observation fidelity loss ($\mathcal{L}_{\text{of}}$), to ensure an accurate learning of SRF. According to the observation model, we degrade the reconstructed and the ground-truth HSIs using the learned SRF and then constrain their disparity with the RGB image. $\mathcal{L}_{\text{dc}}$ and $\mathcal{L}_{\text{of}}$ are defined as

$$\mathcal{L}_{\text{dc}} = \frac{1}{N}\sum_{i=0}^{N}\left\|\hat{\mathbf{R}}^i\hat{\mathbf{X}}^i - \mathbf{Y}^i\right\|_F^2 \tag{25}$$

$$\mathcal{L}_{\text{of}} = \frac{1}{N}\sum_{i=0}^{N}\left\|\hat{\mathbf{R}}^i\mathbf{X}^i - \mathbf{Y}^i\right\|_F^2 \tag{26}$$

where $\mathbf{Y} \in \mathbb{R}^{c\times HW}$ denotes the RGB image, and $\hat{\mathbf{R}} \in \mathbb{R}^{c\times C}$ means the estimated SRF. Thus, the loss function is

$$\mathcal{L} = \mathcal{L}_{\text{re}} + \xi_1\mathcal{L}_{\text{dc}} + \xi_2\mathcal{L}_{\text{of}} \tag{27}$$

where ξ_1 and ξ_2 balance the different terms.

IV. Experiment

A. Experiment Settings

1) Datasets: We carry out experiments on four typical HSI datasets: ICVL [19] dataset, CAVE [54] dataset, and two remote sensing HSI datasets, University of Houston dataset [55] and Chikusei [56] dataset. The ICVL dataset comprises HSIs containing 31 spectral bands, within the spectral range of 400–700 nm with each band spaced 10 nm. All the images are taken outdoors under natural light illumination. 201 HSIs with the size of 1392 × 1300 are included in the ICVL dataset. The CAVE dataset comprises 32 indoor HSIs, each containing 31 bands spanning from 400 to 700 nm, spaced 10 nm apart. Each image is with a size of 512 × 512. The University of Houston dataset contains an image covering 48 bands in the 380–1050-nm spectral range. We remove three bands and use 45 spectral bands. The Chikusei dataset contains an HSI with a spatial size of 2517 × 2335, which is composed of 128 bands in the range of 363–1018 nm. We remove several noise bands and eventually select 100 bands of the images in the experiment.

For the ICVL and CAVE datasets, we select 20 and 24 images, and four and eight images for training and testing, respectively. To test the robustness, we use the SRF of the Nikon D5100 sensor to simulate the training data on CAVE and ICVL datasets, and we adopt the SRF of the Nikon D40 sensor to generate the testing data. For the University of Houston and Chikusei datasets, we take one piece of the image with a size of 512 × 512 for testing, while the remainder serves as training data. The RGB images of HSIs on the University of Houston and Chikusei datasets are simulated using the SRF of Worldview-2 sensor [57].

2) Evaluation Criteria: Five evaluation metrics are employed to gauge the effectiveness of different methods, including peak signal-to-noise ratio (PSNR), spectral angle mapping (SAM), structural similarity index (SSIM), correlation coefficient (CC), and erreur relative globale adimensionnelle de synthèse (ERGAS). The PSNR and CC are used to calculate the difference in spatial quality from the reconstructed HSI to the ground truth. The accuracy of the spectrum reconstruction is shown by the SAM, which is measured along the spectral dimension. The SSIM metric is utilized to quantify the structure similarity in the spatial

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TABLE I
EVALUATION METRICS OF SSR RESULTS OF VARIOUS METHODS ON THE ICVL DATASET

	PSNR(↑)	SSIM(↑)	SAM(↓)	CC(↑)	ERGAS(↓)	Flops(G)	Params(M)	Training Time(h)	Testing Time(s)
HSCNN+ [13] (CVPRW, 2018)	45.01420	0.99507	0.04632	0.99889	6.04752	8380.64	4.65	94.2	11.57
HRnet [12] (CVPRW, 2020)	45.62909	0.99574	0.04352	0.99894	5.72711	4509.08	31.70	94.4	11.70
US3RN [15] (TCI, 2021)	45.50388	0.99456	0.04669	0.99868	6.04966	438.86	**0.20**	**21.0**	15.84
HInet [11] (CVPR, 2021)	38.89735	0.98096	0.10891	0.99477	11.51979	924.25	5.22	36.1	16.10
MST++ [40] (CVPR, 2022)	45.59166	0.99544	0.04741	0.99885	5.77233	641.66	1.62	72.5	11.70
HDnet [17] (CVPR, 2022)	45.17643	0.99472	0.04996	0.99876	6.89296	4784.31	2.67	92.0	**9.50**
DGUNet [58] (CVPR, 2022)	45.40239	0.99496	0.04679	0.99866	5.54020	2919.18	2.12	60.4	26.79
DAUNet [59] (GRSL, 2024)	45.17544	0.99428	0.05082	0.99864	5.95432	4764.85	2.66	90.4	32.27
SSRNet [43](TNNLS, 2024)	45.66202	0.99466	0.04788	0.99726	5.69143	696.22	0.39	30.0	23.39
Ours	**46.30811**	**0.99609**	**0.04215**	**0.99912**	**5.07662**	**377.24**	0.23	23.7	9.88

TABLE II
EVALUATION METRICS OF SSR RESULTS OF VARIOUS METHODS ON THE CAVE DATASET

	PSNR(↑)	SSIM(↑)	SAM(↓)	CC(↑)	ERGAS(↓)	Flops(G)	Params(M)	Training Time(h)	Testing Time(s)
HSCNN+ [13] (CVPRW, 2018)	35.07418	0.97831	0.27584	0.98240	27.50913	1217.79	4.65	12.9	5.99
HRnet [12] (CVPRW, 2020)	34.44442	0.96870	0.25727	0.98620	27.81012	655.22	31.70	34.3	5.56
US3RN [15] (TCI, 2021)	35.50080	0.95887	0.28355	0.98825	23.98156	**66.80**	**0.20**	6.6	10.78
HInet [11] (CVPR, 2021)	30.55521	0.82264	0.45486	0.96378	47.27807	134.30	5.22	6.4	7.37
MST++ [40] (CVPR, 2022)	35.60307	0.95814	0.26354	0.98842	23.12024	93.24	1.61	10.8	5.92
HDnet [17] (CVPR, 2022)	34.69359	0.96238	0.25225	0.98680	26.07776	695.21	2.66	8.7	5.57
DGUNet [58] (CVPR, 2022)	35.45199	0.96856	0.24810	0.98783	24.29669	423.99	2.12	20.9	8.21
DAUNet [59] (GRSL, 2024)	34.87555	0.95988	0.27540	0.98718	25.99886	692.38	2.66	34.7	5.88
SSRNet [43] (TNNLS, 2024)	35.73350	0.95635	0.32735	0.98908	23.10433	101.17	0.39	**5.9**	6.06
Ours	**36.59878**	**0.97932**	**0.22700**	**0.99060**	**20.80393**	150.64	0.22	10.9	**5.11**

domain between two images. The cross-scale image reconstruction errors are calculated by ERGAS as a comprehensive index of image recovery quality. Greater PSNR, SSIM, and CC illustrate better performance, besides lesser SAM and ERGAS. We calculate the number of parameters (Params), floating point operations (FLOPs), and testing time to evaluate the efficiency of different methods.

3) Implementation Details: Before the network training begins, we initialize the penalty parameters μ_1 and μ_2 in (4) and (5) to 1. The step sizes ϵ_1 and ϵ_2 in (10) and (11) are set to 0.03. In this implementation, the parameters balancing the different terms of the loss function, ξ_1 and ξ_2, are set to 1×10^{-4} and 1×10^{-5}. During the experiments, we set the size of paired HSI and RGB samples to 16×16. The batch size is configured to 128, and Adam [60] is employed as the optimizer. We cease the network training at 2500 epochs, the learning rate is initialized to 10^{-4} and halves every 500 epochs. In the PGDL-Net, we set the number of stages to 12. The numbers of the GCN layers and Transformer blocks of our deep proximal operator are set to 1 and 2, respectively. All the compared methods are trained with their original hyperparameters. The experiments are conducted on the PyTorch platform utilizing the NVIDIA 3090Ti GPU.

B. Experimental Results

Nine state-of-the-art (SOTA) methods, including high-resolution dual-domain learning network (HDnet) [17], half instance normalization network (HInet) [11], hierarchical regression network (HRnet) [12], HSCNN+ [13], MST++ [40], unfolding spatiospectral super-resolution network (US3RN) [15], deep generalized unfolding network (DGUNet) [58], degradation-aware unfolding network (DAUNet) [59], and SSRNet [43], are used to compare with our method to test its effectiveness. Each compared method was trained in the same experimental setup to guarantee fairness. The detailed experimental results are as follows.

1) Quantitative Results: The metrics on the ICVL dataset are shown in Table I. The best and second-best metrics are displayed in bold and underlined, respectively. Our algorithm outperforms SOTA methods on all metrics, obtaining the best results on the ICVL dataset. For instance, our algorithm achieves 0.71645-, 0.67902-, and 1.29391-dB improvement in PSNR compared to MST++, HRnet, and HSCNN+. Tables II–IV present the results of different methods on the CAVE, University of Houston, and the Chikusei datasets. The Transformer-based approach yields better recovery than the CNN-based methods. Compared with MST++, the proposed method gets 0.99571-dB improvement in PSNR on the CAVE dataset, and the result on the University of Houston dataset shows a 0.05748-dB rise in PSNR and a 0.03557-dB drop in ERGAS. On the Chikusei dataset, our approach gets a 0.24389-dB promotion compared with the second-best method. With proximal learning for the degradation process of HSI, our algorithm can obtain satisfactory results on data with unseen degradations. Our method has a slightly lower PSNR than SSRNet on the University of Houston dataset, but our method has a better performance in SAM and has a lighter network than SSRNet.

Our method has the lowest FLOPs on ICVL and Chikusei datasets and has the second-lowest Params on all datasets. It means that our method could generally achieve competitive performance with the lighter network. Our method has the second-lowest training time on ICVL and Chikusei datasets. On CAVE and Houston University datasets, the training time of our method is comparable with others. The inferring time is also comparable with other methods, which means that the proposed method could achieve competitive perfor-

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TABLE III
EVALUATION METRICS OF SSR RESULTS OF VARIOUS METHODS ON THE UNIVERSITY OF HOUSTON DATASET

	PSNR(↑)	SSIM(↑)	SAM(↓)	CC(↑)	ERGAS(↓)	Flops(G)	Params(M)	Training Time(h)	Testing Time(s)
HSCNN+ [13] (CVPRW, 2018)	55.83294	0.99891	0.01990	0.99948	2.41575	1221.93	4.66	20.1	3.97
HRnet [12] (CVPRW, 2020)	55.75887	0.99888	0.02028	0.99947	2.40068	659.59	32.05	21.1	3.44
US3RN [15] (TCI, 2021)	55.62533	0.99887	0.02089	0.99946	2.46639	**140.70**	**0.42**	**7.3**	10.69
HInet [11] (CVPR, 2021)	55.03499	0.99866	0.02206	0.99937	2.62408	282.92	11.00	8.7	3.92
MST++ [40] (CVPR, 2022)	55.42685	0.99880	0.02077	0.99943	2.50157	193.42	3.38	25.3	3.80
HDnet [17] (CVPR, 2022)	55.89792	0.99888	0.01968	0.99948	2.36800	696.91	2.67	34.0	3.72
DGUNet [58] (CVPR, 2022)	54.67156	0.99866	0.02302	0.99933	2.76541	434.38	2.14	21.7	2.81
DAUNet [59] (GRSL, 2024)	55.08669	0.99871	0.02307	0.99938	2.66447	696.81	2.67	14.2	**2.76**
SSRNet [43] (TNNLS, 2024)	**56.05674**	**0.99898**	0.01983	**0.99950**	**2.32173**	157.18	0.60	9.3	3.48
Ours	55.95540	0.99892	**0.01951**	0.99949	2.33243	284.24	0.43	11.6	3.05

TABLE IV
EVALUATION METRICS OF SSR RESULTS OF VARIOUS METHODS ON THE CHIKUSEI DATASET

	PSNR(↑)	SSIM(↑)	SAM(↓)	CC(↑)	ERGAS(↓)	Flops(G)	Params(M)	Training Time(h)	Testing Time(s)
HSCNN+ [13] (CVPRW, 2018)	36.76829	0.97172	0.06156	0.97744	12.41682	1237.47	4.72	23.7	14.98
HRnet [12] (CVPRW, 2020)	37.60082	0.96837	0.06946	0.97689	12.22988	665.63	31.74	19.6	8.81
US3RN [15] (TCI, 2021)	38.11473	0.96933	0.05934	0.97952	10.89363	694.78	2.10	6.7	**2.00**
HInet [11] (CVPR, 2021)	34.51667	0.94808	0.09261	0.96510	16.16940	1395.51	54.30	9.4	8.78
MST++ [40] (CVPR, 2022)	37.86291	0.96861	0.06442	0.97818	11.44829	935.66	16.46	15.5	13.24
HDnet [17] (CVPR, 2022)	38.16781	0.97176	0.06030	0.97985	11.05297	703.02	2.69	8.4	8.61
DGUNet [58] (CVPR, 2022)	37.01442	0.96147	0.06805	0.97342	12.47489	475.24	2.21	24.9	3.00
DAUNet [59] (GRSL, 2024)	37.40257	0.96344	0.08858	0.97680	13.30624	713.70	2.74	44.2	2.88
SSRNet [43] (TNNLS, 2024)	37.65491	0.97105	0.06684	0.97630	11.51753	438.11	**1.68**	9.1	2.98
Ours	**38.41170**	**0.97077**	**0.05896**	**0.98118**	**10.66969**	**437.04**	1.74	7.8	6.94

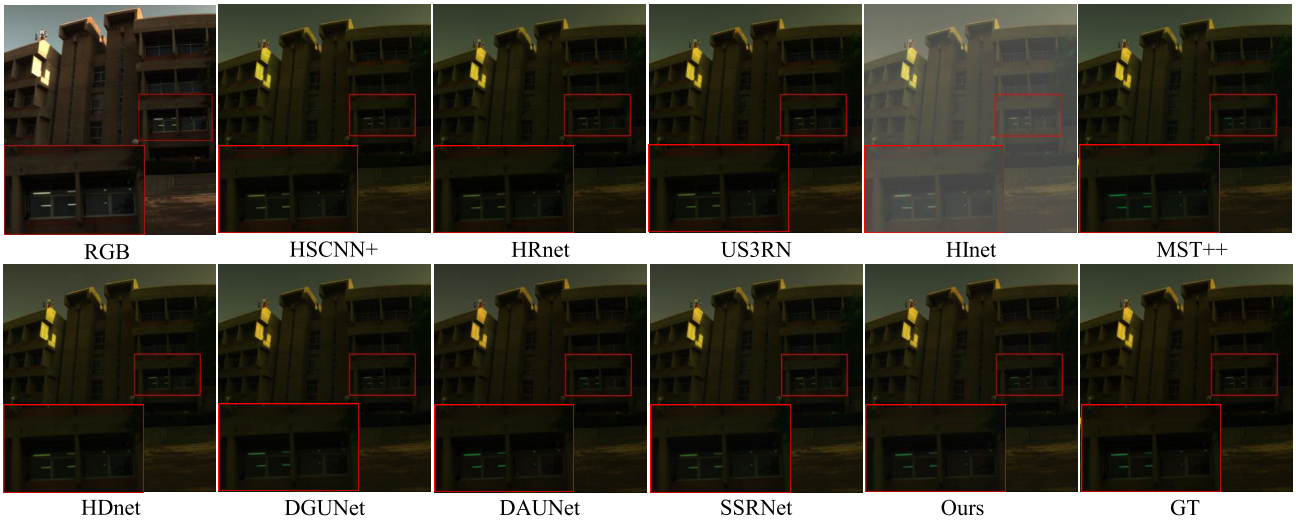


Fig. 3. Reconstructed HSI comparisons of the false color composites simulated by 29th, 18th, and fourth spectral bands on the ICVL dataset. Nine SOTA algorithms and our method are included.

mance with comparable computation compared with SOTA methods.

2) Qualitative and Visual Results: Figs. 3–6 compare the false-color composites simulated by three spectral bands of the reconstructed HSIs with other SOTA algorithms. The RGB image input is shown in the top-left. It is noted that we focus on reconstructing HSI from RGB image, the reconstructed HSI has the same spatial resolution with RGB image, so most of the compared methods could preserve the detailed structures, but the spectral distortion (visual color distortion) appears in the reconstructed HSIs. For instance, in Fig. 4, there is irregular artifacts appearing on the face of the statue in the compared methods. Moreover, in Fig. 6, the reconstructed images of the compared methods have an inappropriate brightness. In contrast, our approach, owing to the integration of both local and non-local features in spatial and spectral domains, performs better in clearer structure and higher spectral fidelity.

The remote sensing scenarios may involve complex background, and the visual difference among different HSIs may be not significant. Thus, in order to highlight their difference, we display the residual error maps (absolute difference between reconstructed and ground-truth HSIs) on four datasets in Figs. 7–10. The darker color and the fewer bright spots indicate that the HSIs recovered by our method are much closer to the ground truth.

In Figs. 11 and 12, we depict the spectral difference curves of the restored HSIs. The difference between the ground truth

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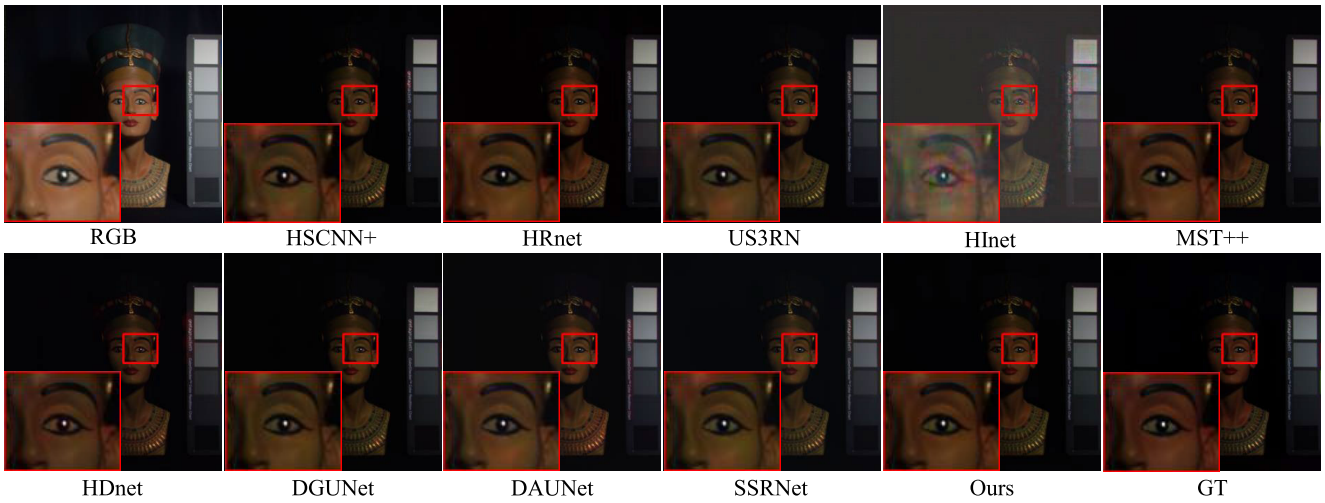


Fig. 4. Reconstructed HSI comparisons of the false-color composites simulated by 29th, 18th, and third spectral bands on the CAVE dataset. Nine SOTA algorithms and our method are included.

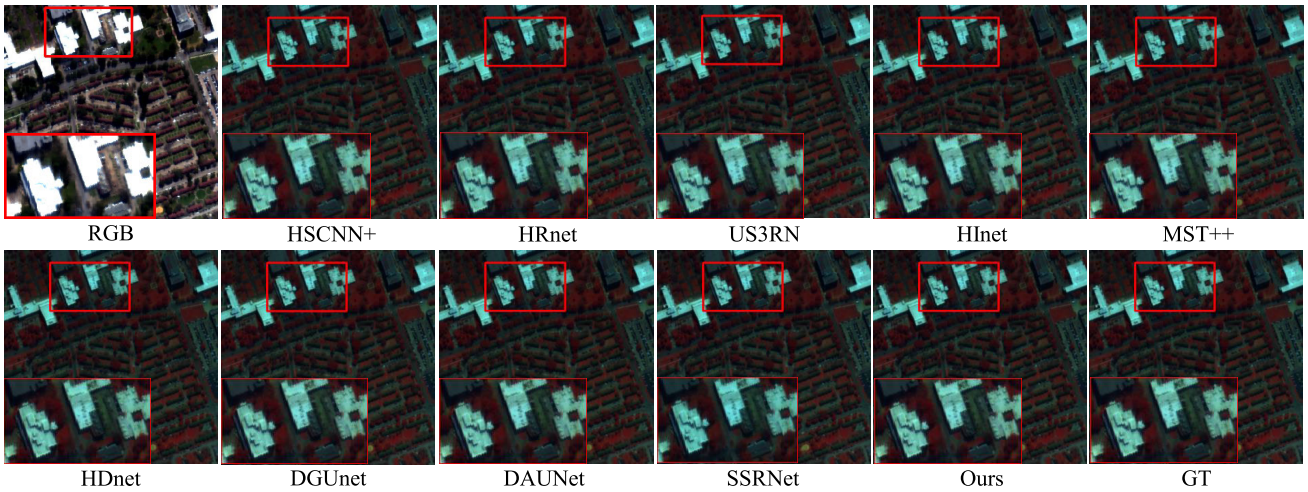


Fig. 5. Reconstructed HSI comparisons of the false-color composites simulated by 41st, 16th, and fourth spectral bands on the University of Houston dataset. Nine SOTA algorithms and our method are included.

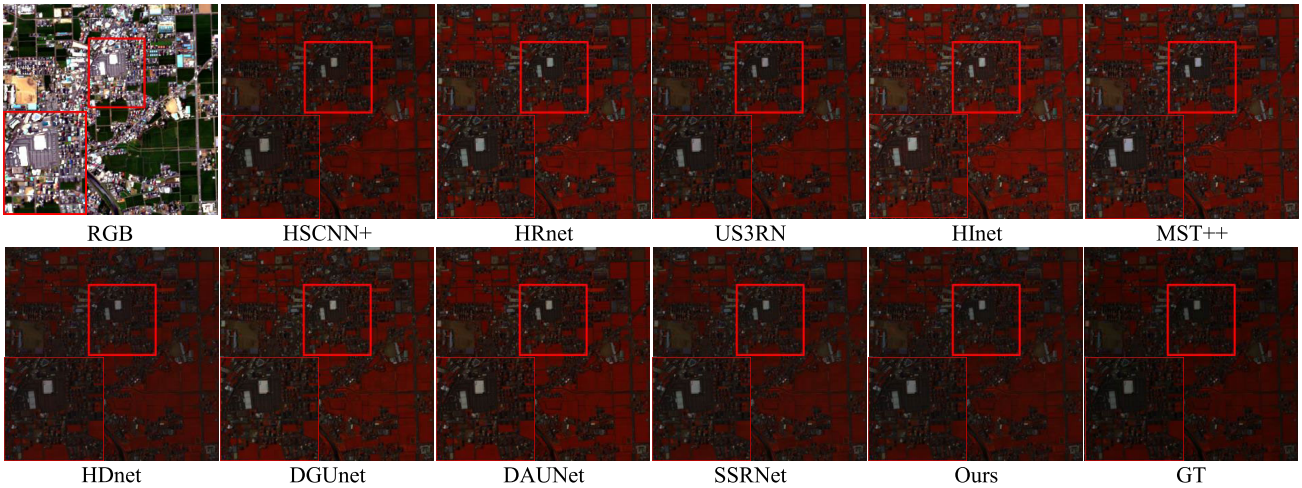


Fig. 6. Reconstructed HSI comparisons of the false-color composites simulated by 71st, 31st, and sixth spectral bands on the Chikusei dataset. Nine SOTA algorithms and our method are included.

and the reconstructed HSIs is indicated by the curve of each point in the image. These visualizations demonstrate that our method results in fewer residual errors, which are closer to the ground truth.

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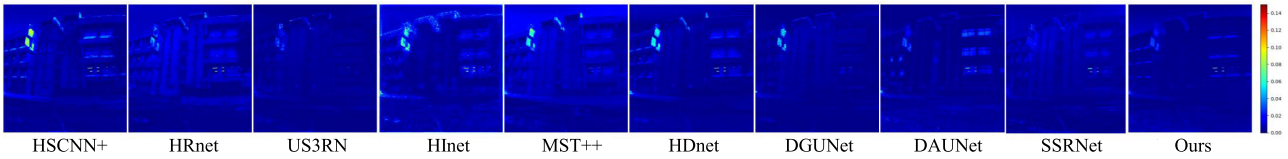


Fig. 7. Error map reflecting the absolute difference comparisons of the 28th band on the ICVL dataset. Nine SOTA algorithms and our method are included.

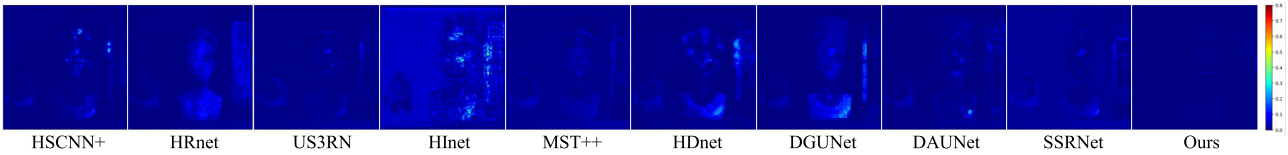


Fig. 8. Error map reflecting the absolute difference comparisons of the 31st band on the CAVE dataset. Nine SOTA algorithms and our method are included.

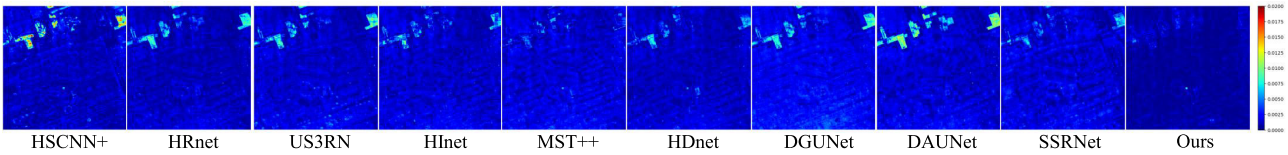


Fig. 9. Error map reflecting the absolute difference comparisons of the first band on the University of Houston dataset. Nine SOTA algorithms and our method are included.

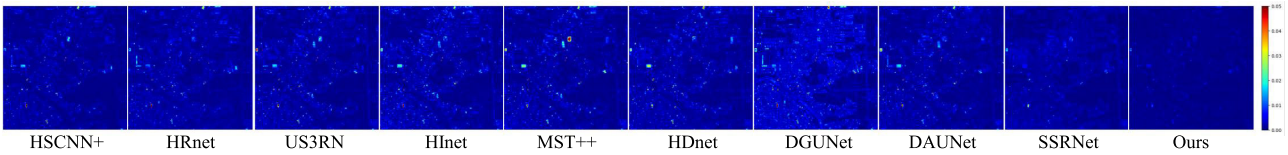


Fig. 10. Error map reflecting the absolute difference comparisons of the eighth band on the Chikusei dataset. Nine SOTA algorithms and our method are included.

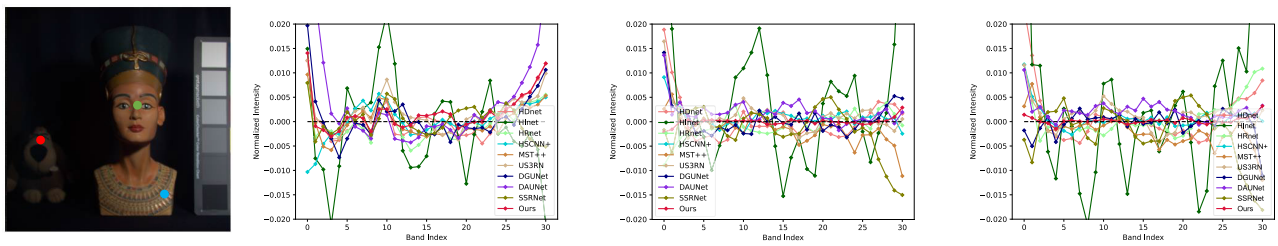


Fig. 11. Comparisons of spectral difference curves on several pixels in the reconstructed HSI on the CAVE dataset. (a) RGB image, from (b)–(d) are the spectral difference curves on (b) red pixel, (c) green pixel, and (d) blue pixel.

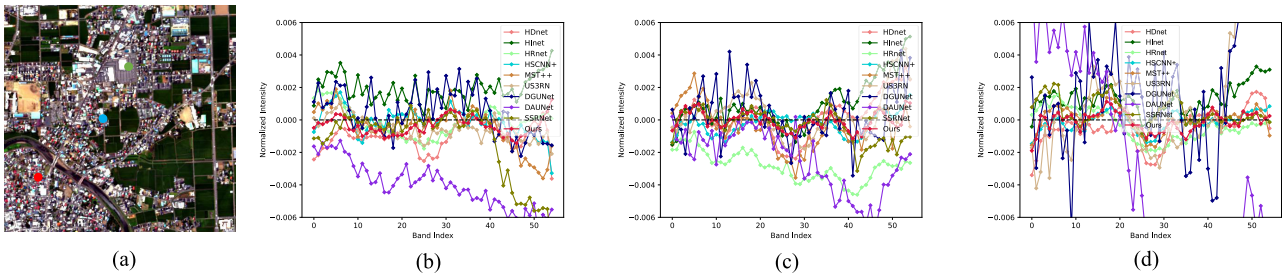


Fig. 12. Comparisons of spectral difference curves on several pixels in the reconstructed HSI on the Chikusei dataset. (a) RGB image, from (b)–(d) are the spectral difference curves on (b) red pixel, (c) green pixel, and (d) blue pixel.

We visualize the feature maps of SISn, GTSn, and PGSn in the deep proximal operator of HSI to illustrate the validity. In Fig. 13, from top to bottom are the outputs of stage 1, 5, and 10. As the number of iterations rises, the quality of reconstructed image gradually improves. Our deep unfolding network learns the different features through three

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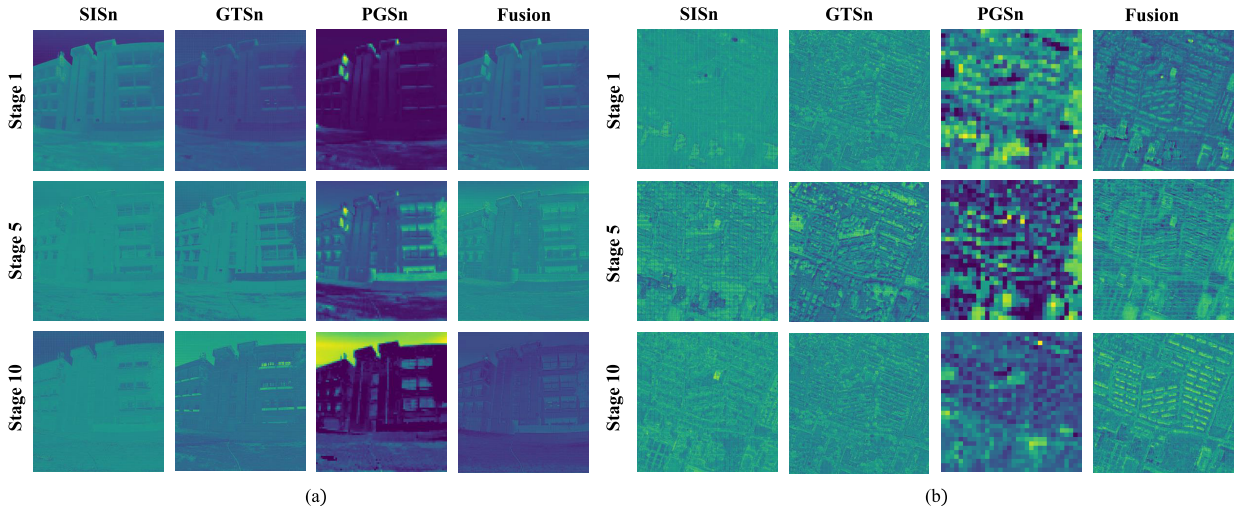


Fig. 13. Feature maps of three modules of our proximal operator. (From top to bottom) Outputs of stages 1, 5, and 10. All the samples are in the third band of each stage. (a) Sample from the ICVL dataset. (b) Sample from the CAVE dataset.

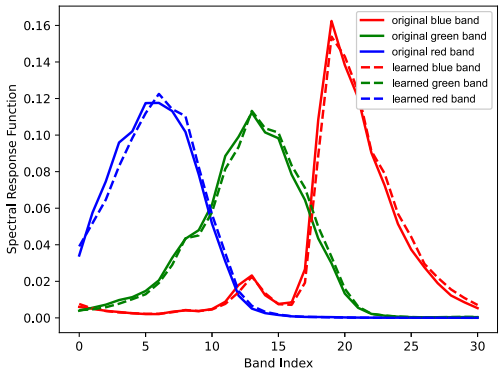


Fig. 14. Visualization of learned and original SRFs.

sub-networks. A better-reconstructed image will be obtained by fusing different modules. The feature maps obtained by fusing features from two branches are more elaborate than the previous features. Fig. 14 compares the learned SRFs and the ground truth. The learned SRFs exhibit similarity with the ground truth and they can capture the primary patterns, for instance, the spectral characteristics over a given range of wavelengths or the wavelength coverages of various bands.

C. Ablation Analysis

We analyze our joint learning, proximal operator for HSI, the loss function, the position embedding in the Transformer, updating $\mathbf{Y}$ and stage number in this section. All the ablation experiments are conducted on the ICVL dataset.

1) Joint Learning: We verify the effectiveness of joint learning the HSI and SRF by setting SRF fixed. As shown in Table V, we present the experimental results of learning HSI and SRF jointly and learning HSI solely. We remove the updating modules of SRF [corresponding to the learning process in (11) and (13)], and only update the learning process of HSI, then analyze the SSR performance. In Table V, it can be found that the SSR network with learning both HSI and SRF would perform better than that of only learning HSI, which means that jointly learning HSI and SRF makes the network more flexible and is thus helpful for SSR.

TABLE V
EVALUATION METRICS OF THE PROPOSED METHOD WITH/WITHOUT LEARNING OF SRF

	PSNR(↑)	SSIM(↑)	SAM(↓)
w	46.3081	0.9961	0.0422
w/o	45.2755	0.9944	0.0501

TABLE VI
ABLATION STUDY OF DIFFERENT PROXIMAL OPERATORS

Resnet	U-net	ST	GTSn	SISn	PGSn	PSNR(↑)	SSIM(↑)
✔	✗	✗	✗	✗	✗	36.9995	0.9733
✗	✔	✗	✗	✗	✗	39.6340	0.9822
✗	✗	✗	✔	✗	✗	44.3495	0.9925
✗	✗	✗	✗	✔	✔	44.1076	0.9920
✗	✗	✗	✔	✔	✗	45.0807	0.9937
✗	✗	✔	✔	✔	✗	45.7853	0.9942
✗	✗	✗	✔	✔	✔	46.3081	0.9961

2) Proximal Operator: We test the effectiveness of our hybrid graph Transformer and analyze its individual module. The results are listed in Table VI. Using GTSn as a proximal operator gives a performance boost to that of Resnet or U-net. Meanwhile, adding SISn and PGSn leads to 0.7312- and 1.2274-dB enhancement in PSNR, respectively. This is attributed to the ability to take both local and non-local information in the spatial domain into consideration and model the long-range correlation along the spectral dimensions. Moreover, to verify the effect of PGSN, we replace PGSN with a spatial-wise Transformer. The experimental result shows that using PGSN leads to a 0.5528-dB improvement in PSNR compared to spatial-wise Transformer. The proposed hybrid graph Transformer could not only explore the spatial long-range dependence as spatial-wise Transformers but also could focus more on the local contextual correlation via the neighboring aggregation of GCN.

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TABLE VII
ABLATION STUDY OF LOSS TERMS

	$\mathcal{L}_{re}$	$\mathcal{L}_{re}+\mathcal{L}_{dc}$	$\mathcal{L}_{re}+\mathcal{L}_{of}$	$\mathcal{L}_{re}+\mathcal{L}_{dc}+\mathcal{L}_{of}$
PSNR(↑)	44.4581	45.9171	45.5578	46.3081
SSIM(↑)	0.9901	0.9949	0.9933	0.9961
SAM(↓)	0.0523	0.457	0.0514	0.0422

TABLE VIII
EVALUATION METRICS OF THE PROPOSED METHOD WITH GTSN WITH/WITHOUT POSITION EMBEDDING

	PSNR(↑)	SSIM(↑)	SAM(↓)
w	46.3081	0.9961	0.0422
w/o	45.8329	0.9950	0.0433

TABLE IX
EVALUATION METRICS OF THE PROPOSED METHOD WITH/ WITHOUT 3×3 CONVOLUTIONAL LAYER FOR $\mathbf{Y}_s$

	PSNR(↑)	SSIM(↑)	SAM(↓)
w	46.3081	0.9961	0.0422
w/o	45.8087	0.9948	0.0469

TABLE X
ABLATION STUDY OF STAGE NUMBER

	9	10	11	12	13
PSNR(↑)	45.9516	42.8405	43.5660	46.3081	44.4901
SSIM(↑)	0.9952	0.9943	0.9918	0.9961	0.9921
SAM(↓)	0.0435	0.0571	0.0525	0.0422	0.0543

3) Loss Function: We verify the efficiency of loss function on ICVL dataset. As shown in Table VII, with the proposed $\mathcal{L}_{\text{dc}}$ and $\mathcal{L}_{\text{of}}$, the network has a better performance in the PSNR, SSIM, and SAM. Adding degradation consistency loss $\mathcal{L}_{\text{dc}}$ to reconstruction loss leads to an increase in PSNR by 1.459 dB and adding observation fidelity loss $\mathcal{L}_{\text{of}}$ to it makes a 1.0997-dB rise. Adding both loss terms, the network gets a 1.881-dB enhancement in PSNR and a 0.0101-dB drop in SAM. It should be noted that in this process we use our graph Transformer as the proximal operator and the stage number is set to 12 in the verification.

4) Position Embedding and Updating $\mathbf{Y}$*:* In Tables VIII and IX, we test the effectiveness of position embedding in Transformer and 3×3 *conv.* layer for updating $\mathbf{Y}_s$, respectively. With these two strategies, the SSR performance is improved by 0.4752 and 0.4994 dB in PSNR, respectively. The main purpose of designing the position embedding is to retain the original relative position information of pixels during the self-attention calculation. We think that the spatial position information of images is significant for SSR. Therefore, we add the position embedding in the Transformer to ensure that image location information can be preserved. Although it is not mathematically necessary to update $\mathbf{Y}_s$ in the iterative process, in the deep unfolding network, the variables in each iterative solution would be the learned deep features. We replace the fixed $\mathbf{Y}$ with an iterative updated $\mathbf{Y}_s$ through a *conv.* layer, which makes the network more flexible and robust.

5) Stage Number: We analyze the stage number S of our algorithm by setting it to different values. As listed in Table X, the performance of our model reaches its peak when the number of stages is 12, so in the implementation, we set 12 stages in our SSR model.

V. CONCLUSION

We present a novel physical guided deep prior learning network for SSR. Considering the physical degradation gives deep learning a higher interpretability, which helps recover HSI. Meanwhile, we design a new proximal operator that combines the GCN and Transformer, providing an effective way to solve HSI. Considering the degradation process from HSI to RGB, we obtain SRF explicitly through unfolding the optimization algorithm and training the unfolding network with an optimized loss function, which effectively enhances the accuracy of reconstruction. In the future, more efficient proximal operators exploiting other priors of HSI can be considered for SSR.

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